

ANNEX

GUIDELINES FOR THE PREPARATION OF THE REPORTS FROM MEMBER STATES ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

I. INTRODUCTION

These Guidelines are intended to assist Member States in the preparation of the reports on the implementation of the 2021 Recommendation on Open Science (hereinafter referred to as “the 2021 Recommendation”) that was adopted by the 41st session of the General Conference of UNESCO on 23 November 2021.

The 2021 Recommendation proposes seven areas of action: (i) promoting a common understanding of open science, associated benefits and challenges, as well as diverse paths to open science; (ii) developing an enabling policy environment for open science; (iii) investing in open science infrastructures and services; (iv) investing in human resources, training, education, digital literacy and capacity building for open science; (v) fostering a culture of open science and aligning incentives for open science; (vi) promoting innovative approaches for open science at different stages of the scientific process; (vii) promoting international and multi-stakeholder cooperation in the context of open science and with view to reducing digital, technological and knowledge gaps.

For the purpose of the 2021 Recommendation, open science is defined as an inclusive construct that combines various movements and practices aiming to make multilingual scientific knowledge openly available, accessible and reusable for everyone, to increase scientific collaborations and sharing of information for the benefits of science and society, and to open the processes of scientific knowledge creation, evaluation and communication to societal actors beyond the traditional scientific community. It comprises all scientific disciplines and aspects of scholarly practices, including basic and applied sciences, natural and social sciences and the humanities, and it builds on the following key pillars: open scientific knowledge, open science infrastructures, science communication, open engagement of societal actors and open dialogue with other knowledge systems.

Pursuant to Articles 15 and 16.1 of the Rules of Procedure concerning recommendations to Member States and international conventions covered by the terms of Article IV, paragraph 4, of the UNESCO Constitution, the Director-General of UNESCO has invited Member States by the Circular Letter (CL/4381) to submit the 2021 Recommendation to their competent authorities within a period of one year from the close of the session of the General Conference at which it was adopted, i. e. before 24 November 2022.

Furthermore, under Article VIII of UNESCO’s Constitution, Member States are required to submit a report on the action taken upon the conventions and recommendations adopted by the General Conference.

II. WHAT ARE THE AIMS OF THIS CONSULTATION?

This global consultation aims to assist Member States in: (1) mapping policies, mechanisms and actions related to open science and all its key pillars, against the objectives of this Recommendation; (2) collecting and disseminating progress and good practices on open science; (3) identifying challenges and opportunities faced by Member States in the implementation of the Recommendation, so as to identify specific capacity-building needs.

III. HOW TO FILL IN THE QUESTIONNAIRE?

The following questionnaire, which was developed with inputs from UNESCO working group on open science monitoring framework, aims to guide and assist Member States with their reporting on the progress made in the implementation of the 2021 Recommendation. It aims to collect information on the extent to which Member States have integrated the core values and guiding principles of open science that are defined in the 2021 Recommendation, in their national science, technology and innovation systems. Responses to this questionnaire will be considered as the official national report of each Member State.

Prior to completing the questionnaire, Member States are encouraged to organize the necessary consultations within and outside the concerned ministries and institutions, including with authorities and bodies responsible for science, technology and innovation, and consult relevant actors concerned with open science, including the scientific community, professional associations, civil society, indigenous and traditional knowledge holders, private sector partners and National Commissions for UNESCO.

In the preparation of their reports, Member States are also encouraged to consult the UNESCO Open Science Toolkit and the first edition of the *UNESCO Open Science Outlook* (published December 2023) containing a collection of information and communication materials relevant to the implementation of the 2021 Recommendation on Open Science. These resources are available online at <https://www.unesco.org/en/open-science/toolkit> and <https://unesdoc.unesco.org/ark:/48223/pf0000387324>.

Member States are requested to designate a contact person responsible for information sharing and cooperation with UNESCO in relation to reporting on the 2021 Recommendation.

Member States are encouraged to submit the questionnaire (in English or French) in one of the following ways:

- (i) Online (preferred): the consolidated questionnaire (one per Member State) can be completed and submitted online.
- (ii) Email: the questionnaire can be completed electronically and sent to: openscience@unesco.org

The responses to the questionnaire, if sent by email, should not exceed 15 pages, excluding annexes and is to be submitted to UNESCO in electronic form only (standard .doc, .pdf or .rtf format).

The reports will be made available on UNESCO's website in order to facilitate the exchange of information relating to the promotion and implementation of this Recommendation.

IV. QUESTIONNAIRE

A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- **Country***: Canada
- **Organization(s) or entity(s) responsible for the preparation of the report***
Name: Environment and Climate Change Canada on behalf of Canada, at the request of Global Affairs Canada and PESCO
Website: <https://www.canada.ca/en/environment-climate-change.html>
Please describe the role/mandate of your organization:
Environment and Climate Change Canada is the lead federal department for a wide range of environmental issues. The department addresses these issues through various actions including the implementation of the Pan-Canadian Framework on Clean Growth and Climate Change; engaging with our strategic partners including provinces, territories and Indigenous Peoples; monitoring; science-based research; policy and regulatory development; and, through the enforcement of environmental laws. The department's programs focus on minimizing threats to Canadians and their environment from pollution; equipping Canadians to make informed decisions on weather, water and climate conditions; and conserving and restoring Canada's natural environment. (From: [Mandate and role: Who we are and what we do - Canada.ca](#))
- **Officially designated contact person(s) who completed this survey:**
Full name*: Alison Kennedy
Position*: Director, Science Management Division, Science and Technology Branch, Environment and Climate Change Canada
FOR ANY FOLLOW UP, please contact:
- **Designated official(s) certifying the report**: Assistant Deputy Minister, Science and Technology Branch, Environment and Climate Change Canada
- **Brief description of the consultation process established for the preparation of the report:**
Environment and Climate Change Canada (ECCC) requested input from Canadian federal science-based departments and agencies, provinces and territories, civil society organizations with expertise in open science, and academia. In addition, as a member of Canada's Permanent Delegation to UNESCO (PESCO), the Province of Quebec contributed a comprehensive response to the questionnaire.
- **Other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey:**

Federal Department or Agency	Website
Agriculture and Agri-Food Canada	agriculture.canada.ca
Canadian Space Agency	www.asc-csa.gc.ca
Canadian Food Inspection Agency	inspection.canada.ca
Canadian Institutes of Health Research	cihr-irsc.gc.ca
Defence Research and Development Canada	www.canada.ca/en/defence-research-development.html
Fisheries and Oceans Canada	www.dfo-mpo.gc.ca

Health Canada	www.canada.ca/en/health-canada.html
Innovation, Science and Economic Development Canada	ised-isde.canada.ca
International Development Research Centre	idrc-crdi.ca
National Research Council Canada	nrc.canada.ca
Natural Resources Canada	natural-resources.canada.ca
Natural Sciences and Engineering Research Council of Canada	www.nserc-crsng.gc.ca
Office of the Chief Science Advisor	science.gc.ca/site/science/en/office-chief-science-advisor
Parks Canada	parks.canada.ca
Polar Knowledge Canada	www.canada.ca/en/polar-knowledge.html
Public Health Agency of Canada	www.canada.ca/en/public-health.html
Public Services and Procurement Canada*	www.canada.ca/en/public-services-procurement.html
Shared Services Canada	www.canada.ca/en/shared-services.html
Social Science and Humanities Research Council	www.sshrc-crsh.gc.ca
Statistics Canada	www.statcan.gc.ca
Transport Canada	tc.canada.ca
Treasury Board of Canada Secretariat	www.canada.ca/en/treasury-board-secretariat.html
PESCO (Permanent Delegation to UNESCO)	www.international.gc.ca/world-monde/international_relations-relations_internationales/unesco/index.aspx?lang=eng#a2
CCUNESCO (Canadian Commission for UNESCO)	en.ccunesco.ca
Provinces and Territories	
Province of Quebec	www.quebec.ca
Provincial/territorial open government contacts	
Civil Society: Academia	
Universities Canada	univan.ca
Vincent Larivière, UNESCO Chair on Open Science, Université de Montréal	unesco.ebsi.umontreal.ca
Civil Society: Other organizations	
Creative Commons	creativecommons.org
Axelys	www.axelys.ca
Érudit	www.erudit.org

1. QUESTIONNAIRE FOR MEMBER STATES' REPORTING ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

1.1 Has the 2021 [Recommendation](#) been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

YES / NO

If yes, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES.

Canadian federal government departments and agencies were made aware of the 2021 UNESCO Recommendation during its development and when it was finalized. During the development of the 2021 Recommendation, Canada set up a working group composed of key actors in federal open science to inform negotiations of the Recommendation. The Canadian Commission for UNESCO (CCUNESCO), a Canadian federal organization, was also involved in drafting the 2021 Recommendation and provided contributions from civil society (e.g., from the [Open Science Beyond Open Access: For and with communities](#) report).

Information on the 2021 Recommendation's approval was also circulated within the federal government's open science community. It was discussed in venues such as the Deputy Ministers' Roadmap for Open Science Steering Committee, as part of a review of the international open science landscape.

In Quebec, the provincial government is involved in promoting and sharing the 2021 Recommendation through the Fonds de recherche du Québec (FRQ), open data resources available online ([À propos – Données Québec](#)) (in French only), a government open data portal ([Données Québec](#)) (in French only), and the Quebec-wide programming competition known as [HackQC](#) (in French only).

1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES / NO

If yes, which language(s)? YES. English and French

1.3 Have there been awareness raising activities on open science, including all its key elements¹, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: [\(i\) 16.f, 16.g, 16.h](#))

YES / NO

If yes, please provide details on the activities organized or foreseen and links as relevant.

YES.

Numerous open science awareness activities and initiatives have been organized across Canadian federal departments and agencies. In particular, the Open Science Action Plans

from 12 federal departments and agencies, now publicly [available](#), help raise awareness of open science. (see Section 2.5). Other Canadian **federal initiatives** include.

- Environment and Climate Change Canada maintains [internet](#) and intranet sites on open science and has set up an internal working group to increase awareness of and implement open science in the department.
- Natural Resources Canada established an Open Science Committee in 2021 to oversee the department's open science activities. The Open Science Committee meets regularly to discuss, update and share information and ideas on open science issues, and has organized awareness sessions for its scientists and senior management.
- The Canadian Space Agency is involved in organizing and participating in public activities such as hackathons, school STEM initiative, seminars, and grants to share insights and maximize the impact of scientific engagements. Some examples include the annual [Space Apps Challenge hackathon](#) and the [Grant for Space STEM initiatives](#).
- The Public Health Agency of Canada established an internal Open Science Community of Practice to raise awareness of open access publications and communication of citizen science. This has included establishing a website with open science resources and publishing an open science quarterly newsletter. The Public Health Agency of Canada also joined the [ORCID Canada Consortium](#) to promote and measure open scholarly communication.
- Health Canada informs employees of open science-related activities through the intranet, intradepartmental announcements and an annual Health Canada Science Forum. Health Canada also promotes awareness of the Open Access Fund, a departmental fund that ensures scientific publications are open and freely accessible at the time of their publication.
- Ongoing efforts are underway at Fisheries and Oceans Canada to enhance communication and promotion of science externally to stakeholders, Indigenous partners and Canadians. In 2024, Fisheries and Oceans Canada launched a new strategic approach to science engagement, including updated governance and targeted deliverables. Activities included renewing the departmental [website](#) and organizing stakeholder engagement activities.
- The Canadian Food Inspection Agency has actively raised awareness of open science through initiatives such as revising its Scientific Publication Policy to incorporate open science principles and participating in interdepartmental meetings to share insights on its benefits and challenges.

Other activities:

In 2020, **CCUNESCO**, in collaboration with the UNESCO Chair in Community-Based Research and Social Responsibility in Higher Education at the University of Victoria, hosted a nine-part global webinar series entitled [Open Science and the Decolonization of Knowledge](#) to raise awareness of open science and gather perspectives from around the world, including from nearly 20 Indigenous Elders and scholars.

In 2022, **CCUNESCO** partnered with UNESCO to publish a [plain language introduction to the 2021 Recommendation](#), and shared the open-source files with other National Commissions to UNESCO to republish in their respective countries and languages. This brochure outlines

the principles and practices of open science and provides a summary of the UNESCO Recommendation on Open Science as an international framework aimed at reducing the technological and knowledge divide between and within countries. It articulates the need for open science and its potential to accelerate response to the global challenges of our time.

Awareness-raising activities have also been organized in recent years by the **Province of Quebec**. For example, data promotion activities took place during the Autorité des marchés financiers' Semaine de la donnée in 2021. The Commission de l'éthique en science et en technologie published "Éthique hebdo" (weekly ethics) posts about the [benefits and risks of open science in the context of COVID-19](#) (in French only) and about the [benefits and ethical risks of sharing scientific data](#) (in French only) in 2020 and 2022. In 2021, the Commission published a discussion paper on [the use of scientific data by public decision makers in a democratic society and ethical considerations](#) (in French only) that examines the benefits and risks of open science. Lastly, the Commission de l'éthique en science et en technologie presented this work to the Quebec (provincial) government and during international events such as the Philosophy Around the World – Worldwide Philosophical Relay Race that was organized by UNESCO in 2021.

1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open scienceⁱⁱ, in publicly funded research? (ref.: [\(i\) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h](#))

YES / NO

If yes, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so?

YES.

Government of Canada (federal) **departments and agencies** that conduct science and research have implemented actions to incorporate open science principles and values. The 2020 [Roadmap for Open Science](#), developed by the Office of the Chief Science Advisor, provides overarching principles and recommendations to guide federal open science activities in Canada, including publicly funded research. Recommendation #3 of the Roadmap called for Government of Canada (federal) departments and agencies to develop [action plans for open science](#), and 12 federal departments and agencies have published their action plans. These action plans have led to more specific actions within federal departments and agencies to incorporate open science values and principles in their research.

Recent federal departmental science strategies are also aligned with this direction. For example, [Environment and Climate Change Canada's Science Strategy 2024-2029](#), which provides direction for mandate-driven science carried out by the department, emphasizes the importance of transparency in its work, achieved in part through open science. The Public Health Agency of Canada's [Science Strategy](#) (2024) similarly prioritizes open and transparent science communication and collaborative approaches to democratize knowledge, and to make public health evidence accessible to all. The [Health Canada Framework for Science and Research Excellence](#) (2022) also values open science approaches in federal government research.

Additionally, Recommendation #4 in the Roadmap for Open Science called for federal departments and agencies to make federal science articles and publications openly

accessible, and advised departments to use a common approach, shared tools and infrastructure to achieve this recommendation. In response to this recommendation, the [Federal Open Science Repository of Canada \(FOSRC\)](#) was launched in January 2024 to provide a stable shared infrastructure for open sharing of research output produced by participating Government of Canada science-based departments and agencies. Other departments and agencies also have stand-alone repositories for their data and information. More information on the FOSRC and other government depositories can be found in response to question 3.4.

Many departments have also entered into transformative agreements with publishers, either individually or collectively via the [Federal Science Libraries Network](#), (a network of seven Government of Canada science-based department and agency libraries) to make open access publishing more accessible to government researchers.

Government of Canada departments and agencies have also held training sessions and developed resources for employees on open science, including:

- Environment and Climate Change Canada has an intranet site providing resources and training materials to employees to support open science implementation in their research.
- Natural Resources Canada has an internal webpage that is dedicated to promoting open science and open access publishing. NRCan has also established a cross-sectoral working group to provide a report and recommendations for open access publishing.
- Fisheries and Oceans Canada funds internal research through the Competitive Science Research Fund (CSRF). All projects funded through the CSRF must have a credible data management plan in place, which addresses open data principles. The Department also publishes a plain-language public report annually, along with communication and outreach products so that all Canadians are informed on the state of the Atlantic, Arctic and Pacific Oceans. The report is based on available scientific information, a large portion of which is publicly funded.
- The Public Health Agency of Canada publishes two Diamond Open Access peer-reviewed scientific journals, the [Canada Communicable Disease Report](#) and [Health Promotion and Chronic Disease Prevention in Canada: Research, Policy and Practice](#).
- Health Canada and the Public Health Agency of Canada both administer departmental Open Access Funds that support researchers and scientists by covering 100% of the charges paid to scientific journals to ensure that federal health publications are open and freely accessible at the time of their publication.
- Statistics Canada makes its publications openly available and free of charge on the [Statistics Canada website](#). Access to Statistics Canada research is further enabled through the [Statistics Canada Open Licence](#), royalty-free, non-exclusive licence to its publications, tables, graphs and reports.

Government of Canada departments and agencies that fund research carried out by external parties have also made efforts to incorporate open science values. Examples include:

- The Canadian Space Agency's grant and contribution funding programs actively encourage recipients to publish their research findings in alignment with open science

principles, including the [smartEarth](#) program¹ and the [Flights and Fieldwork for the Advancement of Science and Technology \(FAST 2021\)](#)² program.

- Fisheries and Oceans Canada provides funding to external science organizations through transfer payment programs. Funding recipients are required to report on data dissemination and how the results were made available to the public.
- The Canadian Institutes of Health Research (CIHR), the Natural Sciences and Engineering Research Council of Canada (NSERC), and the Social Sciences and Humanities Research Council of Canada (SSHRC) (collectively known as the agencies or Tri-Agency) are federal research granting agencies that have developed policies that aim to make the research they fund as open as possible. Specifically, the [Tri-Agency Open Access Policy on Publications](#) (2015) aims to improve access to research funded by those agencies by outlining open access expectations for grant recipients. All peer-reviewed journal publications from Agency-supported research are expected to be freely accessible within 12 months of publication. The agencies have also [recently announced a revision of the 2015 Policy](#) that aims to make agency-supported research freely available at the time of publication. The agencies are engaging with a wide range of partners within the research community to identify and aim to align with national and international funders. A draft of the policy will be released for feedback in the first half of 2025, while the final policy will be released by the end of 2025.
- The [Tri-Agency Research Data Management Policy](#) outlines expectations for research institutions and grant recipients on how research data will be managed and shared. The implementation of the policy requirements will support the core values and guiding principles of open science, including quality and integrity, collective benefit, transparency, scrutiny, critique and reproducibility, and collaboration, participation, and inclusion:
 - In 2025, the agencies will expand pilot projects that require grant applicants to develop and submit Data Management Plans (DMPs) with application materials. The DMP requirement encourages applicants to think about openness of research data.
 - The agencies will also be developing a data deposit requirement, following extensive consultations. Although the data deposit requirement is not an open data requirement, it confirms the agencies' expectations that researchers should provide appropriate access to the data where ethical, cultural, legal and commercial requirements allow, and in accordance with the FAIR principles and the standards of their disciplines.
 - The policy also emphasizes the importance of Indigenous data sovereignty and research data management principles that recognize and respect self-determination for First Nations, Inuit, and Métis Peoples through a distinctions-based approach. The policy has several requirements of researchers engaged with Indigenous

¹ smartEarth is a funding initiative aimed at developing Earth observation applications. It promotes the smart use of satellite data to create solutions for critical global challenges.

² the FAST program supports research projects at Canadian universities and post-secondary institutions that advance scientific knowledge and space technologies, while also providing students with hands-on experience in space-related missions.

research to discourage the unfair and inequitable extraction of data and knowledge. The agencies will continue to engage with Indigenous partners in 2025 to support Indigenous data sovereignty.

The Quebec (provincial) government has implemented specific measures to ensure that publicly funded research reflects the values and principles of open science, notably through the Fonds de recherche du Québec (FRQ) [Common General Rules](#), which were updated in 2024. In addition, the FRQ adopted a [revised version of its Open Access Dissemination Policy](#) in 2022 (initial policy 2019) to strengthen its support for open science and sharing of scientific findings. The FRQ has a requirement to immediately provide open access under an open licence for the work it funds, the findings of which are published in scholarly journals. The FRQ has also set up two programs for fiscal year 2025–2026, a [program to support French scientific journals](#) (in French only) and a [program to support the creation of two French scientific journals in health, science and engineering](#) (in French only) to help provide immediate open access and support the implementation of the Diamond model and open licences. Finally, the FRQ funds the [Réseau québécois de recherche et de mutualisation pour les revues scientifiques \(2024–2025\)](#) (in French only), which aims to support the transition to the Diamond model throughout the Quebec publishing ecosystem.

The Québec provincial government also provides guidance to support knowledge transfer in certain sectors ([Guide to support the implementation of promising practices in an institution](#) (in French only); [Reference framework for knowledge transfer infrastructures in the health and social services network](#) (in French only); [Guide to implementing supporting infrastructure for knowledge implementation and transfer](#) (in French only)).

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 Does your country have a national policyⁱⁱⁱ, strategy or plan of action on science, technology and innovation (STI)?

- does it include any of the following open science elements: open access to scientific knowledge
- open science infrastructure
- open engagement of societal actors
- open dialogue with other knowledge systems

YES / NO

If yes please attach/upload a machine-readable PDF file of the policy and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES.

The Government of Canada developed a [Science, Technology, and Innovation Strategy](#) in 2014. Innovation, Science and Economic Development Canada (a federal department) promotes open science as a key part of this Strategy with actions and commitments intended to promote open access to federal science and publicly funded research. These commitments

were incorporated into the [National Action Plan on Open Government](#). Many federal science-based departments have also developed science strategies (examples in 1.4).

In 2024, the Government of Canada committed to creating an advisory Council on Science and Innovation to better help guide research priorities moving forward. This Council will be made up of leaders from the academic, industry, and not-for-profit sectors, and be responsible for a national science and innovation strategy to guide priority setting and increase the impact of these significant federal investments.

The province of Quebec also has a strategy on research and innovation. More information about the 2022–2027 Quebec Research and Innovation Investment Strategy, including its open science elements, is provided in Appendix 1. Other provinces may also have science and technology-related strategies or action plans.

2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)^{iv} at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) [16.a](#), [16.b](#), [16.c](#), [16.d](#), [16.e](#), [16.h](#))

YES / PARTLY / NO

If yes or partly, please attach/upload machine-readable PDF file(s) of the policy(ies) and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.

YES.

Canada has several national policies and frameworks that include considerations for open science in Canada, including open data.

The Government of Canada joined the international Open Government Partnership in 2011. Membership requires the development of action plans on open government. Canada's resulting [Third](#), [Fourth](#) and [Fifth](#) National Action Plans contained commitments to promote open science, including ensuring that federal science and scientific data are available from open data infrastructure sources, and measuring progress in implementing open science and the benefits it can provide. Canada also developed a [Directive on Open Government](#) (effective 2014), which promotes information management practices that enable the proactive and ongoing release of federal government information to allow for greater transparency, accountability, and engagement. The [Treasury Board Secretariat of Canada](#) has overall responsibility for the Directive; however, all federal departments are responsible for its implementation.

A commitment in the Fourth (2018-2020) National Action Plan on Open Government led to the 2020 [Roadmap for Open Science](#). The Roadmap was informed by the work of the Open Science Roadmap Advisory Committee, which had expert members from within the federal government as well as institutes of higher education. It builds on the 2015 [Tri-Agency Open Access Policy on Publications](#) (see below) and the [Tri-Agency Statement of Principles on Digital Data Management](#), which aims to make research results as accessible as possible. The Roadmap provides overarching principles and recommendations to guide open science

activities in Canada. The recommendations are intended for science and research funded by federal government departments and agencies, both within and outside government.

Canada's [Office of the Chief Science Advisor](#) is the overall lead for open science in the federal government; however, all federal science-based departments are responsible for implementing the Roadmap. As discussed in sections 1.3 and 1.4, a result of the Roadmap, was the development of [open science action plans](#) by federal science-based departments and agencies. Currently, there are 12 plans from the following departments and agencies: Agriculture and Agri-Food Canada, Canadian Food Inspection Agency, Canadian Space Agency, Department of National Defence/Canadian Armed Forces, Environment and Climate Change Canada, Fisheries and Oceans Canada, Health Canada, Communications Research Centre Canada (Innovation, Science and Economic Development Canada), National Research Council, Natural Resources Canada, Public Health Agency of Canada, Statistics Canada. Some organizations are already working on updates to their initial action plans; for example, the Canadian Space Agency's Open Science Action Plan for 2025–2027 is in development.

The [Framework for Implementing Open-by-Default with Federal Government Science](#) (2021), developed to support the Roadmap, provides guidance to federal government scientists in implementing open science. As well, as part of Canada's approach to open science, a committee of experts (the [Advisory Committee on Canadian Scientific Data Governance](#)) was formed in 2023 to provide advice to the Chief Science Advisor of Canada on developing a scientific data governance framework.

The Canadian Space Agency signed the Artemis Accords on outer space exploration on behalf of Canada in October 2020. Signatories of the Artemis Accords pledge to openly share scientific data and publicly release research results.

The Canadian Institutes of Health Research ([CIHR](#)), the Natural Sciences and Engineering Research Council of Canada ([NSERC](#)) and the Social Sciences and Humanities Research Council of Canada ([SSHRC](#)) are federal research funding granting agencies that promote and support research, research training and innovation in Canada. They developed the [Tri-Agency Open Access Policy on Publications](#) (2015, currently under review). The objective of the policy is to improve access to the results of Agency-funded research and to increase the dissemination and exchange of research results (more details in 1.4).

The [2023–2026 Data Strategy for the Federal Public Service - Canada.ca](#) prioritizes open and responsible data flow to improve service to Canadians. It also acknowledges the importance of ethical data stewardship and respect for privacy and data security. Canada's first Data Strategy Roadmap for the Federal Public Services was published in 2018. The Treasury Board of Canada Secretariat has overall responsibility for the Strategy; however, all federal departments are responsible for its implementation; some departments have developed their own data strategies (see e.g., [Health Canada Data Strategy \(2019\)](#)).

In addition to the policies directly related to open government and open science, Canada's [Model Policy on Scientific Integrity](#) (released 2018) fosters a culture that supports and promotes scientific integrity in the design, conduct, management, review and communication of research, science, and related activities in the federal government. The Policy recognizes the importance of openness and transparency in all elements of the research and scientific process and requires the timely release of scientific and research information, in keeping with the Directive on Open Government. All federal departments and agencies with a minimum number of scientific staff are required to develop policies on scientific integrity tailored to their

department, based on the model policy (see [Environment and Climate Change Canada](#), [Fisheries and Oceans Canada](#), [Health Canada and the Public Health Agency of Canada](#), and [Natural Resources Canada](#) for examples of departmental policies). The Office of the Chief Science Advisor of Canada tracks progress in implementing scientific integrity policies in relevant departments and agencies. As of December 2023, all 24 of the departments who were initially required to develop policies had developed them, with 22 policies in effect at that date.

Canada has a House of Commons [Standing Committee on Science and Research](#) that reviews and reports on matters relating to science and research. This committee serves to broadly foster transparency around science and research carried out in Canada. The committee has recently reported on the use of Indigenous Knowledge in public policy development (2024) as well as the role and contribution of citizen scientists (2023).

2.3 In your country, are there specific policy instruments^v that aim to promote open science?

YES/ UNDER DEVELOPMENT/NO

If yes or under development, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

YES.

There are many policy instruments, programs, strategies or internal mechanisms in Canada that promote open science. Federal government examples are described under questions 1.4, 2.2, 2.5, 2.6 and 3.4.

In addition, some of the Quebec (provincial) government's strategic plans and sector action plans support open science by including measures to develop or enhance knowledge and make it available by disseminating new datasets, such as the [SIGÉOM \(Système d'information géominière\)](#) (in French only). Another example is the [Policy Framework for Managing Information Assets](#) (in French only), which lays the foundation for structured and strategic data management within the Autorité des marchés financiers (Quebec's financial market regulatory body). This policy establishes the roles and responsibilities of the Head of Data and Advanced Analytics, who is responsible for setting up data governance and development structures to encourage a data-driven organizational culture. There is also a data development management framework, which was updated in 2024 and aims to maximize the use of data to support decision making; ensure the quality, accessibility and protection of data; and promote the ethical and harmonized use of data across different entities while balancing the benefits and risks of data development.

2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: [\(ii\) 17.a, 17.b, 17.c, 17.f, 17.h](#))

YES / UNDER DEVELOPMENT /NO

If yes or under development, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

YES.

The 2020 [Roadmap for Open Science](#), developed by the Office of the Chief Science Advisor of Canada, is intended to guide open science activities in Canada. Details on the Roadmap and related action plans are described under question 2.2.

In addition, the main objective of the *Fonds de recherche du Québec*'s [Open Access Dissemination Policy](#), adopted in 2019 and revised in 2022, is to promote open access to peer-reviewed publications and to ensure that they are disseminated quickly and available for free (for readers) to the scientific community, potential users, policy-makers and the general public.

2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at the institutional level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.g)

YES /UNDER DEVELOPMENT/ NO

If yes, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science**Error! Bookmark not defined.** included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

YES.

Federal government science-based departments and agencies developed open science action plans in response to the 2020 federal [Roadmap for Open Science](#); departmental scientific integrity policies also serve to promote open science. More detail can be found under question 2.2.

The Quebec (provincial) government has implemented several initiatives that are aligned with the elements of open science at the institutional level, as highlighted in the version of the *Fonds de recherche du Québec*'s [Open Access Dissemination Policy revised in 2022](#). In 2020, [Quebec joined the Open Government Partnership](#) (in French only), an international initiative that encourages departments and agencies to be more transparent and to adopt practices aligned with the principles of open science..

Between 2019 and 2023, the Quebec [Government Digital Transformation Strategy](#) (in French only) focused on reusing and developing data to support economic development and create new value-added digital services in various sectors. The [Government Cybersecurity and Digital Strategy, 2024–2028](#) (in French only), states that public organizations be provided with government guidance on how to access open data. Other initiatives include the 2024–2025 funding program for a Quebec network of scientific journals ([Réseau québécois de recherche et de mutualisation pour les revues scientifiques](#), in French only), *Fonds de recherche du Québec*'s [Open Access Dissemination Policy](#), adopted in 2019 and revised in 2022, which covers the open access dissemination of works published in scholarly journals and master's and doctoral theses. As well, initiatives by other organizations within the Quebec government promote the implementation of open science. Of note is the Quebec Department of Natural Resources and Forests' Data Dissemination Policy (in French only), which provides for the free and open re-use of data from the Department under a Creative Commons licence. This policy supports Quebec's commitment to open government, which aims to promote transparency and free access to data.

The Université de Montréal has the first [UNESCO Chair on Open Science](#) and an associated advisory group on the creation and dissemination of scientific information in French.

The [Tanenbaum Open Science Institute \(TOSI\)](#), part of the Neuro (Montreal Neurological Institute-Hospital) at McGill University, was founded to drive the adoption of Open Science across Canadian neuroscience research institutes. TOSI supports and guides other institutes in adopting Open Science practices and principles and established an alliance of neuroscience research institutes across Canada committed to Open Science.

The [UNESCO Chair in Community Based Research and Social Responsibility in Higher Education](#) based at the University of Victoria and the Society of Participatory Research in Asia (PRIA) supports North–South–South and South–South partnerships that build on and enhance the emerging consensus in knowledge democracy. The UNESCO Chair had an important role in CCUNESCO's outputs on Open Science and decolonization of knowledge in higher education, including a global webinar series (see section 1.3).

2.6 In your country, are there specific funding mechanisms^{vi} for open science at the national level? (ref.: (ii) 17.i, (iii) 18.i, (iv) 19.c)

YES / UNDER DEVELOPMENT / NO

If yes or under development, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

YES.

Examples of funding mechanisms available in Canada are provided here and in response to question 2.7. As well, some of the repositories described in section 3.4 are intended to support open access and replace the need for specific funding mechanisms.

As federal examples, Health Canada and the Public Health Agency of Canada both administer departmental Open Access Funds that support researchers and scientists by covering 100% of the charges paid to scientific journals to ensure that federal health publications are open and freely accessible at the time of their publication.

Examples at the provincial level:

In 2021, the Fonds de recherche du Québec within the Quebec (provincial) government joined [cOAlition S](#), an international group promoting the open dissemination of scientific findings.

At the sector level, the Quebec (provincial) government funds projects led by universities and colleges that involve open science technologies, allowing knowledge to be acquired and shared within a collaborative framework. The [Partnership for Open Access \(POA\)](#), created by [Érudit](#) and the [Canadian Research Knowledge Network](#) 10 years ago, is driving the growth of open access publishing in Canada. It provides financial support to non-commercial scholarly journals through the ongoing commitment of partner libraries and is based on a close collaboration between all stakeholders in the research ecosystem. The POA is based on an open access model in which no fees are charged to readers and authors, known as Diamond open access, thus building a collective, equitable and strong path for knowledge dissemination. Every year, the POA allows more than 2,000 scholarly articles to be published through Diamond open access. The POA financially supports over 250 non-commercial scholarly journals. More than 160 of these journals publish in immediate open access, including 40 that

have made the transition to open access thanks to the POA. The POA is supported by over 90 libraries worldwide.

2.7 In your country, have open science practices^{vii} been integrated in existing research funding mechanisms?

YES/NO

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES.

The (federal) Canadian Space Agency [grant and contribution funding](#) programs actively encourage recipients to publish their results in alignment with open science principles. Examples include the Canadian Space Agency's smartEarth initiative and the Flights and Fieldwork for the Advancement of Science and Technology (FAST 2021) program. smartEarth is the Canadian Space Agency's revitalized funding initiative aimed at developing Earth observation applications. It promotes the smart use of satellite data to create solutions for critical global challenges. The FAST program supports research projects at Canadian universities and post-secondary institutions that advance scientific knowledge and space technologies, while also providing students with hands-on experience in space-related missions (for further details see [Research Opportunities in Satellite Earth Observation | Canadian Space Agency](#)).

Fisheries and Oceans Canada funds internal research through the Competitive Science Research Fund (CSRF) which requires projects to have a credible data management plan in place that addresses open data principles.

The Genomics Research and Development Initiative ([GRDI](#)), which funds genomic research in federal government laboratories in areas such as health care, food safety and global food security, sound management of natural resources, a sustainable and competitive agriculture sector, and environment protection, has projects in eight federal science departments and agencies. The GRDI maximizes access to federally funded scientific research and complies with the federal Open Government Directive.

The [Tri-Agency Open Access Policy on Publications](#) (2015) is being updated by Canada's federal research granting agencies, with the goal of requiring that any peer-reviewed journal publications arising from agency-supported research be freely available, without subscription or fee, at the time of publication. The renewed policy will be released by the end of 2025 (see section 1.4).

Examples at the provincial level:

The Quebec (provincial) government has integrated open science practices into the work it funds. Through its [Common General Rules](#) and [Open Access Dissemination Policy](#), the Fonds de recherche du Québec requires that the work it funds be made openly accessible and published under an open licence. In addition, some of its support programs for French-language scientific journals and funding for the Réseau québécois de recherche et de mutualisation pour les revues scientifiques actively support the transition to immediate open access and promote more open scientific dissemination.

2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: [\(ii\) 17.i](#))

YES / NO

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES.

The Quebec (provincial) government has made efforts to foster equitable public-private partnerships on open science. These partnerships vary from sector to sector. For example, the expertise of the Consortium de recherche et d'innovation en transformation métallique was established to implement the [Critical and strategic minerals research network](#) (in French only) of the Quebec Departments of Economy, Innovation and Energy and Natural Resources and Forests.

In addition, Quebec set up a research promotion structure for all public research institutions called [Axelys](#). This structure is rooted in the desire to improve the dissemination, mobilization and use of scientific knowledge for the benefit of society. Within this structure, numerous public-private partnerships are set up at the project level to maximize the use of knowledge, while complying with university policies that preserve non-commercial research and teaching rights related to intellectual property resulting from public research.

2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: [\(ii\) 17.j](#), [23.c](#))

YES / UNDER DEVELOPMENT / NO

If yes or under development, please provide details and links as relevant.

YES.

As described previously, the [Roadmap for Open Science](#) contains overarching recommendations to guide open science activities in Canada, and for Government of Canada departments and agencies to develop action plans for open science that include a performance measurement element. [Canada's National Action Plan on Open Government](#) also contains commitments and indicators related to open science. Some indicators related to open science from the 2022–2024 National Action Plan are listed below (see [National Action Plan on Open Government Tracker](#) for complete list, including others related to open science).

- C01.1.1 - Add content and features to the Open Science and Data Platform, using user engagement and feedback, both internal and external to government to set priorities for additions
- C01.3.2 - Yearly reports on progress against existing metrics measuring implementation of open science by Science-Based Departments and Agencies, with a focus on open access metrics and diverse types of open access, to be released in the open

- C05.2.2 - Establish an internal and external stakeholder open data advisory working group to help inform open data priorities, help inform the establishment of public reporting on open data release progress, identify open data users and experts, and report publicly on the work and deliberations of the working group and impact of open data
- C05.2.04 - Consult on and develop service standards for prioritizing and releasing open data and using the portal as part of the open data plan.

In addition, the Federal Science Libraries Network has been preparing annual Open Science Dashboards for each member department since 2022, outlining the progress towards open science goals and specific open science indicators.

At least one federal government department (Natural Resources Canada) includes an open science annual performance measurement indicator in its departmental results framework.

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.a](#))

In 2021, Canada spent 1.86% of GDP on research and development expenditures ([Statistics Canada](#), information released December 22, 2023).

3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.b](#))

In 2022, 94% of Canadians had access to the internet at home. Of the respondents who were aware of their internet speed, 87% reported a download speed of at least 50 Mbps ([Statistics Canada, 2023](#)).

3.3 Are there national research and education networks (NRENs)^{viii} active in the country? (ref.: [\(iii\) 18.c](#))

YES / NO

If yes, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to establish them.

YES.

Canada's National Research and Education Network ([CANARIE](#)) is an ultra high-speed network that connects Canada's researchers, educators and innovators to each other and to global data, technology, and colleagues. The NREN connects research facilities and higher education institutions (i.e., most universities and colleges), as well as some teaching hospitals, libraries, museums, federal research labs, and other research and education organizations.

3.4 Are there national or institutional [open science infrastructures](#)^{ix} in your country? (ref.: [\(iii\)](#) [18.d](#), [18.e](#), [18.k](#), [18.l](#), [18.m](#))

YES / UNDER DEVELOPMENT / NO

If yes or under development, please indicate which type(s) of infrastructure.

For each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES.

- federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage.

The Quebec (provincial) government funds the development of [SecureData4Health](#), a secure cloud infrastructure for the analysis and sharing of genomic and health data based on the creation of a new computing node in Quebec and the enhancement of HPC4Health, which is already in place in the province of Ontario. Together, they will form the core of an interprovincial digital platform, which will eventually be expanded nationally. This initiative involves bioinformaticians; statisticians; data and AI scientists; software developers; and influential biomedical, legal and ethical experts from seven of Canada's leading institutions.

The Quebec (provincial) government also supports the funding of research projects aimed at organizing health data and enhancing research capacity in the health and social services network. This research funding will allow health data infrastructure to be developed in the future (see [press release \[in French only\]](#)).

- community managed infrastructures, protocols and standards, for example those that support biodiversity and engagement with society.

The Creative Commons (CC) is a non-profit organization based in the United States with a Canadian entity whose staff support the CC licences. The CC licences give everyone from individual creators to large institutions a standardized way to grant the public permission to use their creative work under copyright law, including scientific outputs ranging from data, publications and hardware.

The digital infrastructure [Érudit](#) supports open digital publishing and research in the arts, humanities and social sciences in the province of Quebec. Documents can be viewed in various formats (PDF, HTML, ePub) and are indexed in search engines and available from the [Érudit website](#). Érudit fits with several of the categories of infrastructure listed in this section; see Appendix 2 for more details.

- platforms for exchanges and co-creation of knowledge between scientists and society

The [Federal Open Science Repository of Canada \(FOSRC\)](#) provides public, web-based access to federally authored scientific articles and publications from the eight participating

science-based departments and agencies.³ While the Findable, Accessible, Interoperable, and Reusable (FAIR) and Collective Benefit, Authority to Control, Responsibility and Ethics (CARE) principles were considered in its development, no comprehensive assessment of compliance has been conducted. Other publicly accessible federal government repositories include Natural Resource Canada's [Open Science and Technology Repository](#), the [National Research Council Canada Publications Archive](#), and the Canadian Space Agency's [Open data and information](#) website. As well, close to 600,000 science and non-science Canadian federal government publications can be found at [publications.gc.ca](#).

Canada's [Open Government Portal](#) contains thousands of datasets, both scientific and non-scientific, from Canadian government departments and agencies. In addition to the Open Government Portal, the Government of Canada hosts and funds the [Open Science and Data Platform](#) which brings together information to better understand cumulative effects. Launched in 2021 and co-led by two federal departments (Natural Resources Canada and Environment and Climate Change Canada), the Open Science and Data Platform provides Canadians with a single point of access to authoritative scientific publications, geospatial data and regulatory information related to development activities.

The Quebec (provincial) government has set up [Données Québec](#) (in French only), an open data portal created as a result of a collaboration between cities and the Quebec government. The portal features 12 datasets in the following fields: agriculture and food; economy and business; education and research; environment, natural resources and energy; government and finance; infrastructure; law, justice and public security; social policies; health; society and culture; tourism, sports and recreation; and transportation.

In addition, interactive tools, online platforms and sector databases benefit from provincial government support or funding, such as [Auchamp.ca](#) (in French only), [RECUPOM](#) (in French only), [SAqE pesticides](#) (in French only), [IRIS phytoprotection](#) (in French only) and [Production fruitière intégrée](#) (the apple producer network's dynamic web platform, in French only).

Additionally, in November 2003, the Institut de la statistique du Québec (ISQ) put the [Databank of Official Statistics on Quebec](#) (Banque de données des statistiques officielles [BDSO]) online. The BDSO provides reliable, objective information in French about Quebec. It also provides access to official statistics, which are the result of a collaboration between partner departments and agencies in Quebec under the ISQ's coordination. Through the BDSO and the entire ISQ website, the public has easy access to a wide range of data on various aspects of Quebec society, such as the economy, education, health and demographics ([Statistics serving society](#)). The Quebec government also funds the [SecureData4Health](#) initiative and is supporting provincial research initiatives to develop health data platforms that will enhance research capacity in Quebec's health and social services network.

The [Réseau Circé](#) is mobilizing the world of French-language scholarly publishing to provide coordinated support for Quebec's scientific journals. Its primary objective is directly related to open science: promoting free and immediate access to Quebec scientific journals in French for the benefit of the entire Francophonie ("Diamond" model). The Réseau Circé is

³ FOSRC participating departments and agencies: Agriculture and Agri-Food Canada, Canadian Food Inspection Agency, Canadian Space Agency, Environment and Climate Change Canada, Fisheries and Oceans Canada, Health Canada, Public Health Agency of Canada, Transport Canada.

structured around four clusters (1. Journals, 2. Platforms, 3. Libraries 4. Research). The Platforms cluster, under [Érudit](#), will support the implementation of the “Diamond” model and open licences in Quebec; consolidate and sustain the business model of French-language scientific journals for immediate open access to research findings and scientific articles; and contribute to improving the adaptability and interoperability of technological tools used or developed by the various players in scientific publishing in Quebec and abroad, with a focus on open access.

The Tanenbaum Open Science Institute (TOSI) has set up or supported multiple neurological research-related open science platforms, including the Neuro Open Biobank, [OMEGA](#) (a repository of brain imaging data), and the [Canadian Open Neuroscience Platform](#).

- **community-based monitoring and information systems to complement national, regional and global data and information systems.**

The Canadian Integrated Ocean Observing System ([CIOOS - Home](#)) holds ocean-related data from various sources, including government, academia, and non-government organizations. Its regional associations use open-access platforms to share data.

[Health Canada's MedEffect Database](#) (the Canada Vigilance Adverse Reaction Online Database) contains information about suspected adverse reactions (also known as side effects) to health products. Adverse reaction reports are submitted by consumers and health professionals, who submit reports voluntarily, and manufacturers and distributors (also known as market authorization holders), who are required to submit reports according to the Food and Drugs Act.

The new [Canadian Drug and Substance Watch \(CDSW\)](#) combines data from multiple sources, such as drug samples submitted by law enforcement (“drug samples”); wastewater monitoring; and monitoring online forums (“web monitoring”). The CDSW builds on surveillance activities that monitor drugs in Canada. It better informs government organizations, public health officials, law enforcement and border control agencies to develop public health responses in addressing the crisis. These responses prevent the spread of emerging drugs and reduce related harms. With a focus on psychoactive substances, the CDSW differs from drug alert systems, which aim to prevent immediate harm by sharing information about the precise location and appearance of drugs.

Fisheries and Oceans Canada participates in global science data infrastructure initiatives, including the [Ocean Biodiversity Information System](#) (OBIS), a global open access data and information repository on marine biodiversity. Fisheries and Oceans Canada has also been a strong contributor to the [Argo project](#) since 2001, a global ocean climate monitoring system that collects data on water properties. The Department also participates in the U.S. National Center for Biotechnology Information and the International Hydrographic Organization.

Community structures in Quebec include the [Baromètre de l'innovation](#) (in French only), which brings together many indicators associated with research, and [Regard 360](#) (in French only), which aims to democratize access to data and give scientific credibility to local, regional and provincial initiatives aimed at reducing inequalities in health and education. Both are currently housed within non-profit structures.

- Other:

As mentioned under question 2.8, a knowledge use organization, [Axelys](#), was created in Quebec to maximize the societal benefits of public research, which in itself could be seen as a measure to create infrastructure to facilitate access to the use of science.

3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: [\(iii\) 18.e](#), [18.f](#))

YES / NO

YES.

[Érudit](#), founded in Quebec in 1998, is a research infrastructure in the arts, humanities and social sciences, whose activities are rooted in the publication and digital dissemination of Quebec scholarly journals.

As a dissemination platform, Érudit's main languages are English and French; some journals are multilingual and publish in other languages, such as Spanish. Érudit mainly benefits researchers and students but also professionals outside of universities in fields such as teaching, nursing, law, urban planning and translation. The platform receives over 35 million visits a year, 70% of which are from outside Canada. See section 3.4 and Appendix 2 for more details.

3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: [\(iii\) 18.g](#))

YES / NO

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science¹, stage of research, accessibility to all the relevant users in the region or internationally).

YES.

Canada is a supporter and contributor to the 2024 G20-GBIF ([Global Biodiversity Information Facility](#)) facilitation process for specific cataloguing and expansion of open biodiversity databases, which includes the GBIF's Biodiversity Information for Development (BID) program and the Global Registry of Scientific Collections (GRSciColl). The goal of these programs is to increase the availability of biodiversity data and information on scientific collections to support nations to meet national and international objectives.

The work of the UNESCO Chair in Community Based Research and Social Responsibility in Higher Education, and CCUNESCO's global webinar series support North-South collaborations and knowledge exchange. See sections 1.3 and 2.5 for more details.

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: [\(iv\) 19.a](#), [19.c](#))

YES / NO

If **yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES.

Within **federal** government departments that carry out open science, a variety of training, guides and other resources have been made available to scientific staff. Several examples are described below.

- The [Federal Science Libraries Network \(FSLN\)](#) comprising seven Government of Canada science-based departments and agencies, is jointly developing a tool to help FSLN researchers search, browse and select open access journals for publishing that are covered by the FSLN transformative agreements with publishers.
- The [National Research Council Canada's](#) library held an author profile clinic during the summer of 2024 to help researchers create and populate their ORCID records. The results and all accompanying materials were shared with other government departments and agencies to facilitate replication of this activity.
- The National Research Council Canada's (NRC) library built a libguide called "Open Science 101 For Researchers" and delivered a presentation on Open Access Publishing at NRC during Canadian Library Month.
- Environment and Climate Change Canada's Library Services offer scientists and researchers resources and training to facilitate open access of publicly funded research, publications and data.
- The Public Health Agency of Canada has provided training and support to its researchers and authors to support the use of ORCID persistent identifiers in publication, to identify and select Open Access journals, and to contribute their work to the Federal Open Science Repository of Canada. The PHAC Library has also developed reference guides on Open Access publishing, ORCID usage, and mechanisms to offset Open Access publishing costs, including Transformative Agreements.
- Fisheries and Oceans Canada has developed a guide for employees on how to publish articles in open access journals and on how to create and maintain ORCID IDs. In addition, the Department's scientific staff have received training in plain language communication.

In the Quebec (provincial) government, training on the obligations arising from the Fonds de Recherche du Québec (FRQ)'s Open Access Dissemination Policy has been systematically offered to FRQ employees since 2022 and to new grant holders since 2023. Also, the FRQ webpage for the [version of the Open Access Dissemination Policy revised in 2022](#) (in French only) contains frequently asked questions, videos and other resources on open access publishing.

The Quebec University Library Partnership's ([Partenariat des bibliothèques universitaires du Québec](#), in French only) mission is to "equip Quebec universities with innovative information management platforms." Its priority area 3 "Open science and support for research activities" includes the implementation of a shared service offering to support scholarly journals.

The Érudit organization (see section 3.4, Appendix 2) organizes numerous knowledge mobilization activities for its user communities, journal editorial teams and researchers who publish in its journals to help them understand and adopt open access and open science best practices. The Érudit team organized three recent events on these issues: a two-day conference entitled “Perspectives sur la publication et les revues savants : évolution du libre accès, des pratiques d’évaluation et des nouvelles possibilités de recherche” (Perspectives on publishing and scholarly journals: evolution of open access, evaluation practices, and new opportunities for research) held as part of the 2023 annual Acfas⁴ Conference (May 11 and 12, 2023, Université de Montréal); and two symposia for the editorial teams (one for scholarly journals based in Quebec and the other for those based across Canada). Érudit also published a research note on [open science](#) (2021, in French only).

The Tanenbaum Open Science Institute (TOSI) provides support and informational materials on open science to neurological researchers through its annual Neuro Open Science in Action Symposium and the open science educational resources on its [website](#).

The Open Climate Campaign led by Creative Commons, SPARC (Scholarly Publishing and Academic Resources Coalition) and EIFL (Electronic Information for Libraries) provided several workshops and webinars to researchers on open science practices. The Campaign ended in 2024.

4.2 Have capacity building programmes for policy-makers on how to integrate [core values](#) and [principles](#) of open science in STI policies and strategies taken place in your country?

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES.

A federal interdepartmental working group dedicated to developing and implementing open science practices within the Government of Canada has been established, chaired by Environment and Climate Change Canada.. Participants meet bimonthly to share lessons learned, best practices and challenges in implementing open science. As well, some federal departments and agencies have created internal open science networks or working groups. For example, the Public Health Agency of Canada has launched an internal Open Science Community of Practice to enhance awareness of open science practices and supports for researchers, science communicators and policy-makers. Environment and Climate Change Canada has a working group that aims to implement open science across the Department.

In Quebec, a variety of measures have been implemented to develop decision makers’ capacity in both open science and science in general. For example, the [Chief Scientist of Québec](#) advises the Quebec (provincial) government on the state of science in relation to a wide range of societal issues. The Office of the Chief Scientist raises the profile of Quebec research in Canada and around the world and promotes careers in research and scientific culture. In 2024, the Quebec government also announced the [creation of a table of municipal chief science advisors](#) (in French only) that would be a forum for discussion and collaboration

⁴ Acfas was previously known as the Association canadienne-française pour l’avancement des sciences (1923-2001) and Association francophone pour le savoir (2001-2019).

to establish a consistent approach to the practice of municipal scientific advice across Quebec. Quebec has also set up innovation zones. They are designed to attract researchers from Quebec and elsewhere to collaborate on innovation projects in promising technological sectors in which Quebec has significant comparative advantages. These researchers work in close collaboration with municipalities, research and teaching institutions, economic organizations and businesses. Scientific knowledge generation is one of the societal benefits of these innovation zones.

The Érudit organization actively works to inform decision makers at various levels of the principles and values of open science, including within the consortium's universities, libraries and research-funding agencies. To do so, Érudit produces numerous notes and reports for decision makers and organizes events, such as the panel "Towards the adoption of diverse approaches to open access: Promoting the dissemination and use of knowledge across disciplines" held during the 2023 Canadian Science Policy Conference.

4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: [\(iv\) 19.b](#))

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

[no response]

4.4 Are open educational resources^x as defined in the [2019 Recommendation on Open Educational Resources](#), used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: [\(iv\) 19.d](#))

YES/ PARTLY/ NO

YES.

There are open educational resources on open science available in Canada at several universities and through other institutions. For example, the University of Alberta Library's [website on Open Science](#) is licensed for non-commercial use under a Creative Commons license, as are the [open science information materials](#) at the Neuro, McGill University.

4.5 Have there been any initiatives in your country to support science communication in line with open science values and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision makers and the public at large? (ref.: [\(iv\) 19.e](#))

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES.

Government of Canada (federal) departments and agencies share and promote the results of their scientific research to decision makers and the public at large. Federal science is publicly funded science, and driven by the mandate of each department, for the benefit of Canadians.

As well, many federal departments make publications openly available in government repositories (section 3.4), such as the Federal Open Science Repository of Canada (FOSRC), with the FOSRC in the process of implementing plain language summaries to help research reach a broader audience. Federal departments and agencies contribute research data to Canada's [Open Government Portal](#). Other examples of federal government efforts to disseminate scientific knowledge are listed below:

- Environment and Climate Change Canada posts links to published research regularly on social media (X, LinkedIn, Facebook); the department also has various programs aimed at communicating science such as:
 - [Ask a Scientist](#): a video series promoting ECCC's science where Canadians can ask questions to scientists
 - [Science in Action](#): a video series showcasing ECCC's scientists in the field
 - Scientist-at-large: an initiative that connects ECCC scientists with the public through speaking engagement activities to communicate ECCC science in a more meaningful way
- The Canadian Food Inspection Agency (CFIA) shares and publishes scientific information across many channels, including on [inspection.canada.ca](#), on [social media](#), through [statements and news releases](#), on the [Cultivating Science blog](#), and through media interviews. Additionally, scientific reports on food safety are currently [available on the CFIA website](#).
- Parks Canada's science communication initiatives include national reports (e.g., [Parks Canada Carbon Atlas Series – Carbon Dynamics in the Forests of National Parks in Canada](#)), #ParksCanadaConservation communication products via social media (X, Facebook, Instagram, [YouTube](#)) and [Parks Canada's website](#), as well as a variety of interpretation programs and products offered at National Parks, National Historic Sites, National Marine Conservation Areas and the National Urban Park administered by Parks Canada.
- The [Health Canada and Public Health Agency of Canada Scientific Integrity Policy](#) gives researchers and scientists the right, and encourages them, to speak about or otherwise express themselves on science and their research without approval or pre-approval. To this end, Health Canada and the Public Health Agency of Canada developed the [Science and Research Communications Handbook](#).
- The Public Health Agency of Canada and Health Canada jointly publish a blog series ([The Science of Health](#)) to share information with the public on their scientific activities and achievements in plain language.
- To highlight the importance of social and behavioural sciences in public health, the Public Health Agency of Canada provides public information on its [Behavioural Science](#) program and products.
- Fisheries and Oceans Canada publishes scientific advice resulting from its formal peer review process online ([Publications](#)), increasingly with links to supporting data. In this way,

DFO is making government science available to the public by providing information on the scope of the science advice, its intent, the data inputs, and the participants involved in the peer-review.

- Natural Resources Canada publishes scientific briefs and links to scientific information via social media channels, including through its [Simply Science](#) products.
- Statistics Canada has several publications to disseminate scientific knowledge to researchers of different disciplines, decision makers and the public at large including: [Health Reports](#), [Economic and Social Reports](#), [Juristat](#) and [Survey Methodology](#). In addition, all new data and products are communicated to Canadians through [The Daily](#) and the [StatsCAN mobile application](#).
- Grant applications submitted to the Canadian Institutes of Health Research, the Natural Sciences and Engineering Research Council of Canada, and the Social Sciences and Humanities Research Council of Canada must include a knowledge mobilization description that identifies key audiences and how they will be reached.

In partnership with UNESCO, the Canadian Commission for UNESCO published [An introduction to the UNESCO Recommendation on Open Science in March 2023](#). CCUNESCO shared the open source files of this brochure with other National Commissions, encouraging them to republish it in their respective languages. Dissemination of the brochure and CCUNESCO's work to ensure the inclusion of Indigenous perspectives in Open Science dialogues has taken shape in webinars, podcasts, and in-person events, for example, the University of Toronto's Knowledge Equity Lab podcast, [Perspectives on the UNESCO Open Science Recommendation](#).

In Quebec, the Fonds de recherche du Québec (FRQ) aims to make access to scientific knowledge more equitable and to contribute to citizen scientific literacy ([Science ouverte \[in French only\]](#)). The FRQ organizes events on the dissemination of scientific knowledge, such as the panel entitled [Support strategies for the transition to open access](#) during the Canadian Science Policy Centre (CSPC) conference in November 2024. The FRQ also organized a session entitled [Relever le défi de la diffusion en libre accès des publications scientifiques](#) (Rising to the challenge of open access dissemination of scientific publications) held during the 91st Acfas Conference in May 2024. The overall theme of the conference—Mobiliser les savoirs en français (Knowledge mobilization in French)—reflects an effort to share research results with the broader community.

The Quebec (provincial) government publishes certain data for the general public in the form of a directory, such as the directory of innovative wood construction projects ([Répertoire de projets de construction innovants en bois](#), in French only). In collaboration with the university network, the Quebec government organizes free virtual meetings open to all for the transfer of scientific knowledge, such as the [Forest knowledge rendezvous](#) (in French only). These meetings provide a forum for exchanging and sharing expertise.

Sector-specific conferences, such as the annual [Québec Mines + Energy](#) Conference (Appendix 3) organized by the Quebec (provincial) government also provide an opportunity to foster knowledge exchange and knowledge mobilization.

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

- 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: [\(v\)](#) [20.b](#), [20c](#), [20i](#))**

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES.

Federal science-based departments and agencies are increasingly moving toward recognizing and incentivizing open science values and principles in career progression frameworks for their research scientists. See also Section 2.7 which provides information on how open science is considered in research application processes and most of that information would be relevant to this section.

Through the Quebec (provincial) government , the Fonds de recherche du Québec became a member of the Coalition for Advancing Research Assessment (COARA) and plans to review its assessment processes in the near future.

- 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: [\(v\)](#) [20.a](#), [20.c](#), [20.d](#))**

YES / NO

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES.

Federal science-based departments and agencies are increasingly moving toward recognizing and incentivizing open science values and principles in career progression frameworks for their research scientists.

Within the Government of Canada, career progression for federal research scientists is guided by a Career Progression Management Framework for Federal Researchers (2006). The Framework identifies “collaborative research and development projects and public outreach” among the list of productivity outputs expected of researchers for their career advancement. Specific departments may provide further guidance, recognition and incentives related to open science. For example, at Natural Resources Canada, scientists involved in open science activities can submit relevant evidence when presenting their dossiers for evaluation and career advancement.

The Public Health Agency of Canada, with input from other science-based departments and agencies in the Government of Canada, is currently undertaking a review of the criteria for promotion outlined in the Career Progression Management Framework for Federal Researchers, in order to incorporate updated priorities and guiding principles, including open science. Any revisions are not expected to be fully implemented until at least 2026-27, to allow researchers, managers, and review committees time to become familiar with the interpretation and application of any criteria, including open science practices.

Within the Quebec (provincial) government, the Fonds de recherche du Québec has standardized its scientific reporting templates to include requirements for information gathering, open access and open licences under its Open Access Dissemination Policy. In addition, the Fonds de recherche du Québec has adapted its open access publication requirements to the specifics of funding large research groups, which sometimes include hundreds of researchers under the same funding, to allow a gradual transition to immediate open access.

5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: [\(v\)](#) [20.a](#), [20.b](#), [20.c](#), [20.e](#), [20.f](#), [20.g](#), [20h](#))

YES / NO

If yes, for each initiative, please provide more information, including on the stakeholders^{xi} that are involved or lead the initiative, and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES.

The Tanenbaum Open Science Institute (TOSI) offers [Open Science awards and prizes](#) for researchers to promote change in the way scientists are evaluated. The awards focus on qualitative impact rather than the number of publications.

5.4 Have there been any unintended negative consequences^{xii} of open science practices reported in your country? (ref.: [\(v\)](#) [20](#))

YES / NO

If yes, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

YES.

It is not clear if negative trends in scientific publishing, including an increase in predatory publishers and high article processing charges, can be attributed to open science practices. There have been efforts within the federal government science community to mitigate these trends, such as the distribution of information on predatory publishing to scientific staff and through negotiating agreements with publishers to reduce or eliminate article processing charges.

The federal science community is also concerned with the potential negative consequences of open science/open data practices in the context of Reconciliation and the assertion of Ownership, Control, Access and Possession (OCAP) principles by Indigenous communities. As more datasets become widely available, ones that have restrictions on use may be less

likely to be used, particularly with the rise of AI. This may further marginalize perspectives of Indigenous communities.

The Quebec (provincial) government has identified a funding issue in the publication of research findings in scholarly journals with immediate open access. Scholarly journals sometimes charge publication fees, and fees are sometimes required for Diamond open access journals to operate consistently.

Another obstacle is the lack of recognition for open access publishing practices in the standards used to evaluate researchers' scientific output. This prevents the scientific community from adopting this practice. It is also difficult to counter the appeal of publishing in prestigious scholarly journals which do not publish in immediate open access.

There can be perceived conflict between the principle of open science and the need to make knowledge accessible for different audiences. This is in line with the open science values of access and benefits-sharing for society, but misunderstandings sometimes persist among certain scientific or societal stakeholders.

Finally, a lack of positive regional benefits for locally funded or supported science presents a challenge. The free and automatic dissemination of knowledge does not necessarily guarantee that it will be used for the benefit of the community that generated it. This underlines the importance of having stakeholders who are dedicated to mobilizing scientific knowledge in order to encourage its adoption and use. Although this issue does not conflict with the principles of open science, it is recurring, and mechanisms for sharing the societal benefits of scientific output must be developed.

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: [\(vi\) 21\)](#)

YES, at the national level.

YES, at the institutional level.

YES, at both national and institutional levels.

NO

If yes, please indicate which of the recommended actions they address and provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES, at both national and institutional levels.

▪ promotion of pre-prints

Several repositories described in section 3.4 include or have been set up to include pre-prints, such as the [Federal Open Science Repository of Canada \(FOSRC\)](#) and Natural Resources Canada's [Open Science and Technology Repository](#).

- exploration of open peer-reviewed practices
- valuing and sharing of negative scientific results and associated data
- participatory methods that value inputs from societal actors, e.g., citizen science, crowdsourcing scientific projects

Results from CCUNESCO's nine-part global webinar series [Open Science and the Decolonization of Knowledge](#) encouraged the 2021 Recommendation to include participatory and intersectoral science elements to ensure that Indigenous knowledge and perspectives would be included respectfully and with integrity in science activities.

Canada's [Citizen Science Portal](#) is a compilation of citizen science projects across the country. Projects are led by a variety of organizations, including universities, non-government organizations and government departments.

The Canadian Food Inspection Agency has established collaborations with Indigenous communities in Canada. These efforts embody Inclusive Science by respecting Indigenous Knowledge systems and including them in scientific research processes. Such collaborations ensure that diverse perspectives are integrated into the design, execution, and dissemination stages of research, fostering mutual learning and co-creation of knowledge.

The Public Health Agency of Canada has successfully implemented participatory/citizen science methods for disease and vector surveillance via [FluWatchers](#) and the [Tiger Mosquito Citizen Science Study](#).

Health Canada included a commitment in Canada's fifth [National Action Plan on Open Government 2022–2024](#) to support the development of necessary infrastructure, including guidance, capacity building, and information technology, to better enable citizen science more effectively in a health context. To deliver in part on this commitment, Health Canada provides interdepartmental coordination on participatory research and citizen science to benchmark researchers' needs, inform infrastructure requirements and develop guidelines and standards to ensure safe, secure, and responsible collection, management and analysis of participant data and contributions.

Health Canada undertook [public opinion research](#) in 2023, to better understand the public's interest, motivators and barriers to participating in government-led research. With a focus on traditionally underrepresented demographics of the population, including low-income and rural communities, visible minorities, and Indigenous peoples in Canada, this work is informing efforts to develop guidance for researchers seeking to recruit and retain volunteers in federal citizen science research.

In its response to the [2023 study by the Standing Committee on Science and Research](#) on the role and value of citizen scientists in Canada, Health Canada has committed to exploring how to streamline processes, simplify public access, and increase the ease of use of the results in government projects across the federal science environment. For example, Health Canada's partnership with Laboratories Canada is exploring enterprise-wide digital solutions and enablement, and support for reducing barriers to science collaboration in the federal science environment. Health Canada is also working with Shared Services Canada's Science Program to develop a prototype (proof of concept) participatory research exchange platform which would meet Government of Canada standards for accessibility, official languages and security.

Participatory research/citizen science is also reflected in [Health Canada's Framework for Science and Research Excellence](#), under "Innovation and Real-World Learning" underscoring the

importance of citizen science as a key element of the department's toolkit to strengthen Health Canada science and support representative and timely decision making.

The accessibility of science, expressed in terms of knowledge transfer, is now evaluated by all Quebec (provincial) government research-funding agencies. The current trend is to take it one step further with specific programs for knowledge co-creation; action research; partnership research, discussions on commercializing research prior to scientific output; and research focused on addressing major societal challenges. The Fonds de recherche du Québec also funds a number of participatory research programs involving citizen science including [ENGAGE - 2025–2026](#) (in French only).

- development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas

Some Quebec (provincial) government calls for proposals, such as [Concerted action: Cultural safety, experience and needs of First Nations and Inuit users of the health and social services network](#), (in French only) aim for the knowledge generated by the funded project to improve our understanding of First Nations and Inuit people's use of, and experience with, health care and social services, and to generate discussion to support decision making on cultural safety in health and social services establishments.

- promoting open innovation practices
- other: open access publishing by default

Several members of the Federal Science Libraries Network opted for "Open Access by default" in their transformative agreement with the publisher Springer. Open Access by default ensures that all affiliated articles are open and accessible at the time of their publication.

- other: providing funding support

Health Canada and the Public Health Agency of Canada's Open Access Funds support researchers and scientists by covering 100% of the charges paid to scientific journals, to ensure that publications funded by these departments in the federal health portfolio are open and freely accessible at the time of their publication.

- other: new approaches to research assessment

The federal research granting bodies (The Canadian Institutes of Health Research (CIHR), the Natural Sciences and Engineering Research Council of Canada (NSERC), and the Social Sciences and Humanities Research Council of Canada) within the Government of Canada signed the [San Francisco Declaration on Research Assessment \(DORA\)](#), a Declaration that recognizes the need to improve the ways in which the outputs of scholarly research are evaluated. DORA signatories include more than 60 other Canadian universities and research institutions.

- other: approaches to data management

Tri-Agency research data management policy is in line with FAIR and CARE principles. The agencies are involved in ongoing engagements with institutions, organizations and partners to support Indigenous research data management principles.

The Data Management Policy for the Tri-Agencies within the Government of Canada encourages applicants to include data management plans with funding applications in order to encourage data openness and the sharing of research results.

- other: creation of knowledge-mobilization organizations.

In Quebec, the creation of [Axelys](#) (see sections 2.8 and 3.4) is expected to allow a greater number and diversity of stakeholders to benefit from the positive societal impacts of scientific knowledge generation.

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: [\(vii\) 22.a](#))

Canadian federal government departments and agencies are members of many international research organizations that support open science.

Some examples are provided below:

Since 2017, the Canadian Space Agency has partnered with NASA to provide Canadian data, training resources, and the support of experts and judges for the International Space Apps Challenge, the world's largest annual global hackathon. It aims to encourage participants to solve real-world challenges using open data from and/or about space. This partnership has continued ever since, engaging participants from across the country.

In 2020, NASA, in coordination with the U.S. Department of State and seven other initial signatory nations, established the Artemis Accords. The Canadian Space Agency signed the Artemis Accords on behalf of Canada in October 2020. These Accords reinforce the signatories' commitment to key international agreements, including the Outer Space Treaty, the Registration Convention, and the Rescue and Return Agreement, while promoting best practices and responsible behaviour in civil space exploration. Signatories of the Artemis Accords also pledge to openly share scientific data and publicly release research results, ensuring that people worldwide can benefit from the progress made in space exploration and discovery.

Canada supports the United Nations Decade of Ocean Science for Sustainable Development ([Ocean Decade](#)) and Fisheries and Oceans Canada works with the Canadian ocean community, such as the Oceans Research in Canada Alliance (ORCA), to advance ocean science efforts. Fisheries and Oceans Canada engages in several other international initiatives that advance open science, including the North Pacific Marine Science Organization and the International Council for the Exploration of the Sea ([ICES](#)). ICES is an international marine science organization that develops impartial evidence on the state and sustainable use of our seas and oceans. It developed a Transparent Assessment Framework (TAF), an open data resource that aims to make data, methods and results from ICES assessments transparent, easy to find, explore and re-run.

The Quebec (provincial) government has also developed international collaborations on open science. The Fonds de recherche du Québec is a member of cOAlition S, an international consortium of granting agencies that supports immediate open access to scientific publications and implements [Plan S](#).

The UNESCO Chair on Open Science at the Université de Montréal, Professor Vincent Larivière works with 17 partners from 10 countries to advance understanding of Open Science in its various forms, and its impact on the dissemination of scientific knowledge nationally and internationally.

7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: [\(vii\) 22.b, 22.e, 22.f](#))

YES / NO

If yes, please provide links as relevant and more information.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES.

Canada is an active member of the G7 Open Science Working Group, which was established in 2016 during the G7 Ministers of Science and Technology meeting in Tsukuba, Japan. It aims to *share open science policies, explore supportive incentive structures, and identify good practices for promoting increasing access to the results of publicly funded research, including scientific data and publications*⁵.

Please see section 7.1 for other examples of existing collaboration.

7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: [\(vii\) 22.c](#))

YES / NO

If yes, please provide links as relevant and more information.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

NO.

We are not aware of Canada's involvement in any regional (multi-country) or international funding mechanisms for open science. Regional examples of funding mechanisms within Canada are provided in other sections of this document.

⁵ [Communiqué of the G7 Ministers of Science and Technology 2016 \(utoronto.ca\)](#)

8. GENERAL CONSIDERATIONS

8.1 re there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

YES /NO

If yes, please provide more information.

YES.

Institutions in Canada are facing funding challenges to support the development and maintenance of open science practices, such as open access repositories, data-sharing platforms, and training programs.

Ensuring compliance with privacy laws and safeguarding sensitive data, for example in the fields of food safety and Indigenous research, creates additional layers of complexity in implementing open science principles.

Another key challenge is finding a balance between openness and the protection of proprietary research and intellectual property, particularly in collaborations with private industry or sensitive government work. For example, in Quebec much of the Autorité des marchés financiers' data are stored in systems co-owned with other members of the Canadian Securities Administrators (CSA). Access to and use of this data is governed by an intellectual property and licensing agreement for the CSA's national systems that includes controls and restrictions that the various members must comply with. There are also limitations on the open access dissemination of knowledge due to national security requirements, which are playing an increasingly larger role in the Canadian scientific landscape.

Another barrier to full participation in open science is the commercial nature of scientific publishing. While Diamond Open Access venues are available, individual researchers are motivated by prestige and impact associated with scientific journals published by for-profit corporations. Institutions that support open access publication, via payment of article processing charges or transformational subscription agreements, are nevertheless contributing to a publishing model that is not motivated by the principle of knowledge as a common good, and participation in science as an inherent right. New strategies put in place by for-profit journals to counter the anticipated loss of revenue from the increase in immediate open access and open licence publishing (e.g., charging fees to publish with Green open access) impedes implementation of the Recommendation.

APPENDIX 1

Question 2.1

Title: 2022–2027 Quebec Research and Innovation Investment Strategy (QRIIS²)

Authority in charge of developing the strategy: Ministère de l'Économie, de l'Innovation et de l'Énergie (MEIE), Quebec government

Entity involved in the development process : The 2022–2027 QRIIS² was developed through a broad consultation:

- 233 briefs were received and analyzed.
- A virtual regional tour was organized to define the priorities of the Quebec research and innovation (R&I) ecosystem.
- The Grand Rendez-vous de l'innovation québécoise, organized with the support of the Conseil de l'innovation du Québec, convened more than 2,500 participants and invited the Premier and several ministers to speak. It consolidated the consultative process and presented the government's policy directions in the realm of R&I.
- Other groups were consulted, including businesses in Quebec, representatives of strategic groups, college centres for the transfer of technologies (CCTTs), applied research centres, and the thematic networks of the FRQ, as well as secondary school, CEGEP and university students.

Year of adoption: 2022

Year of last update: On October 11, 2024, Perspectives SQRI²: rencontre de mi-parcours was held to take stock of the first two years of QRIIS². The meeting opened a dialogue on the actions to be taken to maximize the positive effects of the second half of the QRIIS²'s implementation.

Link to the press release on Perspectives SQRI²: rencontre de mi-parcours :

<https://www.quebec.ca/nouvelles/actualites/details/strategie-quebecoise-de-recherche-et-dinvestissement-en-innovation-2022-2027-linnovation-pour-creer-de-la-riche-58830> (in French only)

Key areas of the strategy: The 2022–2027 QRIIS² introduces mechanisms and tools to meet the current needs and challenges of stakeholders in the Quebec research and innovation (R&I) ecosystem and relies on their mobilization. The strategy focuses on five key priorities:

- Priority 1: Excel in research, science and technology.
- Priority 2: Create an environment conducive to developing innovation.
- Priority 3: Support in-company investments and the commercialization of innovations.
- Priority 4: Develop talent and a culture of science and innovation.
- Priority 5: Support promising sectors and catalyst projects.

Open access to scientific knowledge: The QRIIS² mentions the issue of access to research findings in the form of scientific publications or raw research data. The FRQ will help provide immediate access to scientific publications, promote the circulation of open data and other research products, and establish open science infrastructure.

Open engagement of societal actors : The QRIIS² mentions that the FRQ will be adding a science and society section to their program on key societal challenges. This section will tackle obstacles to dialogue between science and society, especially disinformation, and will propose solutions to foster dialogue, such as participatory and citizen science.

The Quebec (provincial) government also supports research and innovation through other plans and policies, including the [2023–2027 Strategic Plan](#) (in French only) of the Department of Agriculture, Fisheries and Food, the [Bio-Food Policy 2018–2025](#) (in French only) and the [2022–2025 Quebec Life Sciences Strategy](#).

APPENDIX 2

The digital infrastructure Érudit supports open digital publishing and research in the arts, humanities and social sciences in the province of Quebec. Based on international standards, this digital publishing chain produces thoroughly marked-up digital documents in XML format, which makes the documents easier to access and use as research data. The files produced are optimized for online dissemination (tracking and quick download) and are available in PDF, XML, HTML and ePub formats. Several digital publishing options are offered based on the journal's specific needs. Documents can be viewed in various formats (PDF, HTML, ePub) and indexed in search engines. They are then available from the [Érudit website](#), as well as from several discovery tools, aggregators and search engines. The browsing tools being developed by Érudit (integrated search engine, notices) also help with dissemination.

The textual and digital data generated by Érudit meet most of the FAIR principles. This includes the addition of rich metadata, including persistent identifiers, such as ISSN, DOI, ORCID and ROR (currently being implemented). These metadata are generally machine readable. As a result, they are detected by organizations such as Crossref and the open database OpenAlex.

The majority of communications generated by Érudit, other than published articles, have CC BY⁶ open licences. This is also the case for data generated in-house and made available for research. Lastly, Dataverse and Zenodo platforms have been created on behalf of the organization to make documents and data accessible.

Érudit embraces the Diamond open access model, in which no fees are charged to authors or readers, and 97% of the 162,000 scholarly articles disseminated are open access. To ensure the financial sustainability of open access journals, Érudit partnered with the Canadian Research Knowledge Network to create the Partnership for Open Access (POA) (see question 2.6). The POA's goal is to transform the commercial agreements under which libraries purchased subscriptions to access journal content into a collaboration aimed at financially supporting the journals.

As a result, Érudit owns and manages a vast amount of data, including both the articles' metadata and the articles themselves. The metadata are completely open and gives rise to bibliometric research in science and technology studies and the sociology of science. The vast body of textual data managed by Érudit is also accessible to researchers who request authorization.

Érudit's IT infrastructure is based mainly on free software and open standards (FAIR principles).

⁶ CC BY licences are a type of Creative Commons licence in which licensees are able to "distribute, remix, adapt, and build upon the material in any medium or format, so long as attribution is given to the creator" and allows for commercial use (Source: [Creative Commons website](#)).

APPENDIX 3

The annual [Québec Mines + Energy](#) Conference brings together the mining and energy sectors to showcase cutting-edge knowledge, innovations and stakeholders. By bringing the mining and energy sectors together, the conference provides a unique synergy and includes a high-calibre program recognized for its quality and diversity. The program highlights the intersection between these two sectors; promotes student competitions designed to showcase scientific advances and promote studies in the geosciences and mining; and includes a discovery component in which nearly 2,000 students and teachers learn about the mining and energy sectors and their complementary contribution to the energy transition.

Endnotes (from UNESCO)

-
- i Please refer to the elements of open science defined in the 2021 Recommendation, namely : [open scientific knowledge](#) (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), [open science infrastructures](#), [open engagement of societal actors](#) and [open dialogue with other knowledge systems](#).
- ii The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.
Please see the [values of open science](#), namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.
Please see the [guiding principles for open science](#), namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.
- iii National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision making processes in the area of STI on the national level.
- iv These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.
- v Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).
- vi A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.
- vii Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy-makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.
- viii A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.
- ix Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.
- x Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others ([2019 Recommendation on Open Educational Resources](#))
- xi For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.
- xii For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.